ITA0608-MACHINE LEARNING

LAB PROGRAMS

NAME: SAMUEL I

REG NO:192312584

DATE:14/07/26

EXP NO:1 FIND S-Algorithm

AIM:

To implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis from the given training data

PROCEDURE:

1. Define the training dataset
2. Initialize the hypothesis with the most specific values (0).
3. Consider only the positive training examples
4. Update the hypothesis by replacing mismatched attributes with ?
5. Display the final hypothesis

PROGRAM:

```
data = [
    ['Sunny','Warm','Normal','Strong','Warm','Same','Yes'],
    ['Sunny','Warm','High','Strong','Warm','Same','Yes'],
    ['Rainy','Cold','High','Strong','Warm','Change','No'],
    ['Sunny','Warm','High','Strong','Cool','Change','Yes']
]

h = None

for row in data:
    if row[-1] == 'Yes':
        if h is None:
            h = row[:-1]
        else:
            for i in range(len(h)):
                if h[i] != row[i]:
                    h[i] = '?'

print("Final Hypothesis:", h)
```

OUTPUT:

```
Final Hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

RESULT :

Thus ,the FIND-S algorithm was implemented successfully ,and the most specific hypothesis was obtained .